Mesiri

Field Guide · 03 of 13 · Internal team edition

Materials

What is on site, what came, what went into the work, and where the money disappears when nobody is counting.

4	2	1	Yes
pages working today	pages planned for V1	page planned for Later	works on WhatsApp

Draft, 7 Nov 2026. Construction details here are our understanding and must be checked by someone with site experience before we use them with customers.

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1 · THE PROBLEM WITHOUT IT

Money leaves the company as material, and nobody sees where it goes

> *"Near project closeout, you suddenly discover a shortage of 4 loads of metal. Reflected on paper, but missing on the ground. Accounts are tied to that: if money left, the material should be there."*
>
> GM of a Kerala builder, sales meeting, 6 Nov 2026

Material is usually the biggest cost on a building site, bigger than labour. It arrives in many small loads, from many suppliers, to several sites, and is used a little every day by different teams. On most sites it is tracked in a paper register that reaches the office days later, if at all. These are the problems that follow:

WHAT GOES WRONG	WHAT IT LOOKS LIKE ON SITE	WHO FEELS IT, AND WHAT IT COSTS
Shortage found too late	The slab is planned for Friday; on Thursday evening the engineer finds 30 bags of cement left, not 130.	The team waits or the pour is moved. 6 people idle for a day is about ₹6,400 of wages for nothing, plus an urgent purchase at a worse rate.
Paid for, not on site	The bill says 4 loads of metal; the ground shows less. Short delivery, unloading at another site, or theft.	Accounts pays the full bill. Nobody can prove what arrived, so the loss is never recovered.
Used more than the estimate	Plastering uses 9 bags per 100 sq ft where the estimate said 7. Nobody compares.	The margin on the job shrinks quietly. The QS finds out at the end, when it can't be fixed.
Things you can't count	Sand and metal are dumped in a heap. Once dumped, nobody can count them.	Only "what came" and "what was used" can tell you what should be there. Without both, a heap is a guess.
Double orders and leftovers	Two engineers order cement the same morning. The electrician asks for 10 pipes, uses 6, the rest disappear.	Cash locked in extra stock; leftovers wasted or taken.
Moved between sites, not recorded	20 bags taken to the Chennai site in the company's own lorry.	One site looks short, the other has extra that nobody knows about.
Price creep	Cement is ₹14 a bag dearer than last month from the same shop.	Nobody notices, because last month's rate is in a different notebook.
The office is blind	The MD calls the engineer to ask "how much steel is left?"	Every answer costs a phone call and is only as good as the engineer's memory.

A rough example of what one leak costs. If a load of 20 mm metal is about 200 cft (sizes vary by lorry) at ₹46 a cft, the GM's "4 loads missing" is about **₹36,800**, on one material, on one site. Mesiri's monthly fee is a fraction of that.

What the customer actually wants

Not a digital copy of the paper register. The GM said it plainly: "Just noting down that metal arrived or cement arrived has no real benefit." What he wants is to **know, while there is still time to act**:

- what is on each site right now, and how many days it will last;
- that what was paid for actually arrived;
- that what was used matches the work done (this needs the BOQ, see Guide 07 Cost control);
- a shortage a week before it happens, not at the end of the project.

2 · THE PROBLEMS PEOPLE HIT WHILE USING IT

Why materials software usually dies by week three, and how Mesiri avoids it

Contractors have tried registers, Excel and apps before. Most stop using them. The reasons are almost always the same, and each one is a design rule for us.

WHY IT FAILS	HOW MESIRI HANDLES IT
Too much typing on site. A storekeeper with dusty hands will not fill a 10-field form for every lorry.	A voice note or a line on WhatsApp ("200 bag cement vannu"). Mesiri prepares the entry; the person taps **Save**. Only material and quantity are required.
Units don't add up. Sand comes in loads, is used in cft or bags; steel in tons, used in bars.	Each item has one **stock unit**. Standard conversions are automatic (1 ton = 1000 kg, 1 brass = 100 cft, a cement bag = 50 kg). For a "load", Mesiri asks once, an admin answers ("1 load M-sand = 60 cft"), and it is used for everyone after.
The same item under five names. "M sand", "msand", "manal", "Msand 1st quality".	One item list per company with **aliases**. Mesiri never guesses a "near" material: if it's not on the list, it asks or adds it, it never saves cement as cement paint.
No opening stock. The site already had 60 bags when Mesiri started, so stock is wrong from day one.	**Opening stock** is its own entry type, and can be loaded from Excel at setup.
Only deliveries are recorded, never use. Stock only ever goes up.	Use is one line ("45 bag cement footing-nu"). Planned: comparing use with the BOQ will flag a site that reports deliveries but no use (CC-08).
Nobody trusts the numbers. Someone edited last week's entry; the total changed; no one knows why.	Nothing is edited silently. A correction is a new entry that reverses the old one, with who and why. Every number on Stock can be traced to its entries in Movements.
Price and bill are not known at the gate.	Supplier and rate are optional at delivery; they come later from the bill (planned: bill photo matched to the delivery, PRC-11). A missing price shows as "Not recorded", never as ₹0.
The office screen is confusing. Twenty columns, codes, menus.	Three tabs, each answering one question: **Stock** (what do we have), **Movements** (what happened), **Items** (what we use).

3 · WHO USES IT, AND WHEN

Six people, six different questions

WHO	WHEN	THEIR QUESTION	WHERE THEY GET IT
Site engineer / supervisor	Daily, several times	"Record that cement came." "Is there enough sand for tomorrow?"	WhatsApp
Storekeeper	Every delivery	"Did the lorry bring what the challan says?"	WhatsApp

WHO	WHEN	THEIR QUESTION	WHERE THEY GET IT
Project manager	Daily / weekly	"What is running low? Should I move stock from another site?"	WhatsApp, Stock tab
Purchase / accounts	Weekly	"What came from Malabar this month, at what rate? Is the bill right?"	Movements tab, Finance
Quantity surveyor (QS)	Weekly / monthly	"Does the cement used match the work done?"	Reconciliation (planned)
GM / MD	Weekly	"How much did we spend on materials? Any shortages coming?"	Stock tab, weekly report

The site people should never need the dashboard. If an engineer has to open a laptop to record a delivery, we have already lost.

4 · ONE WEEK ON A REAL SITE

Villa, Kakkanad: the wall putty that went missing

The numbers below are the demo company you will see in the screenshots.

1. **Friday, opening stock.** The storekeeper sends the counts once: 60 bags cement, 300 cft M-sand, 400 cft metal, 25 bags wall putty, and the rest. Mesiri records them as opening stock.
2. **Sunday to Tuesday, deliveries.** 200 bags cement from Malabar Hardware at ₹400, M-sand and metal from Periyar Sands, TMT steel from Kerala Steels. Each one is a WhatsApp line, confirmed with Save.
3. **Monday to Thursday, use.** "45 bags cement, column footings grid B." "650 blocks, ground floor walls." Each use names the work it went into.
4. **Thursday, a delivery with no price.** 40 bags of wall putty arrive; nobody knows the rate yet. Mesiri saves it without a price and marks spend as incomplete instead of pretending it was free.
5. **Friday morning, the gap shows.** The site reports 70 bags of putty used for the interior. Only 65 ever arrived (25 + 40). Stock shows **−5 bags, Out**, and a "running low" banner appears at the top of the page.
6. **Friday, 10 minutes later.** The PM asks the storekeeper on WhatsApp. A 10-bag delivery on Wednesday was never sent. It is recorded with its real date; putty is now +5, and the week's numbers are right.

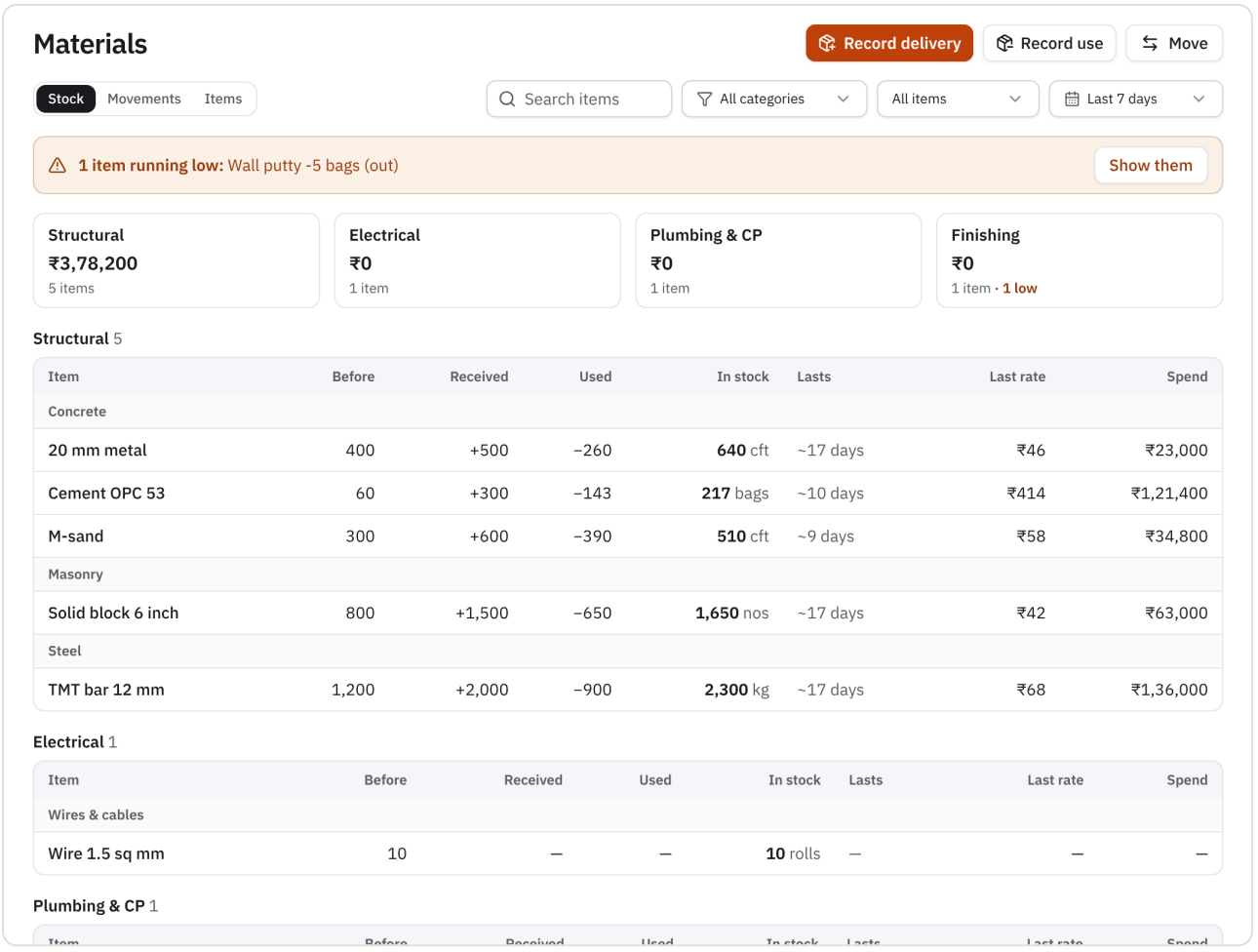
The same gap on paper would have surfaced at month end, if ever. That is the whole value of the module in one story: **a mismatch becomes visible the same day, while someone still remembers what happened.**

5 · THE PAGES, AND HOW THEY LOOK

Three tabs today, three pages to come

Stock Working

Answers: **what do we have on site right now, and how long will it last?**



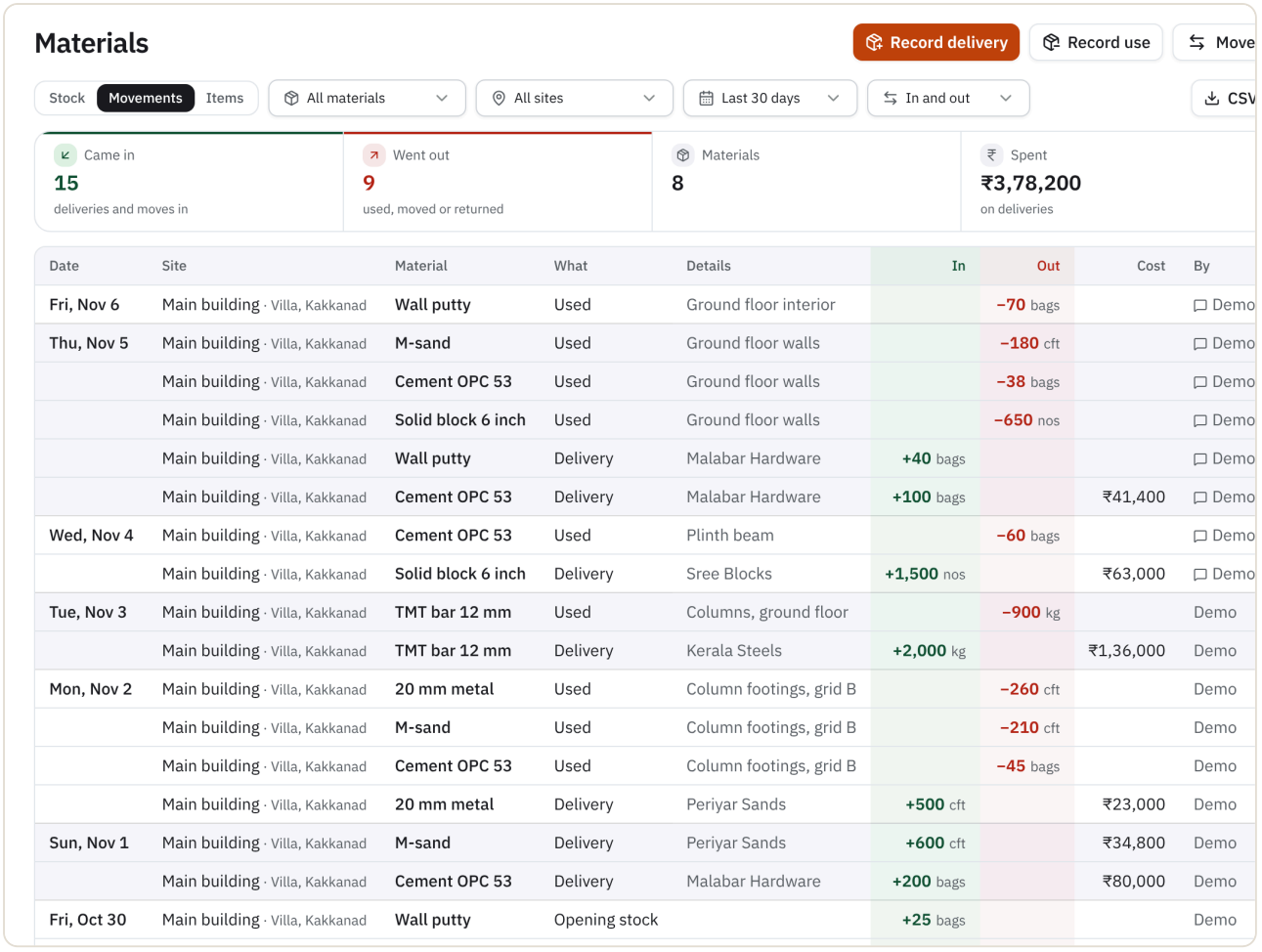
Stock tab, last 7 days, all projects. Demo data.

What must be on it, and why:

- **Before → Received → Used → In stock** in one row, so anyone can check the sum in their head (60 + 300 − 143 = 217).
- **Lasts** (days left at the recent rate of use): the number that turns a register into a warning.
- **Running low** banner on top, and negative stock in red. A problem must be impossible to miss.
- **Spend by category** (Structural, Electrical, Plumbing, Finishing) for the GM, and **Last rate** for purchase.
- Filters: period, project/site (from the top bar), category, search by any name.

Movements Working

Answers: **what happened, when, who reported it, and what was it for?** This is the register, the record every Stock number comes from.



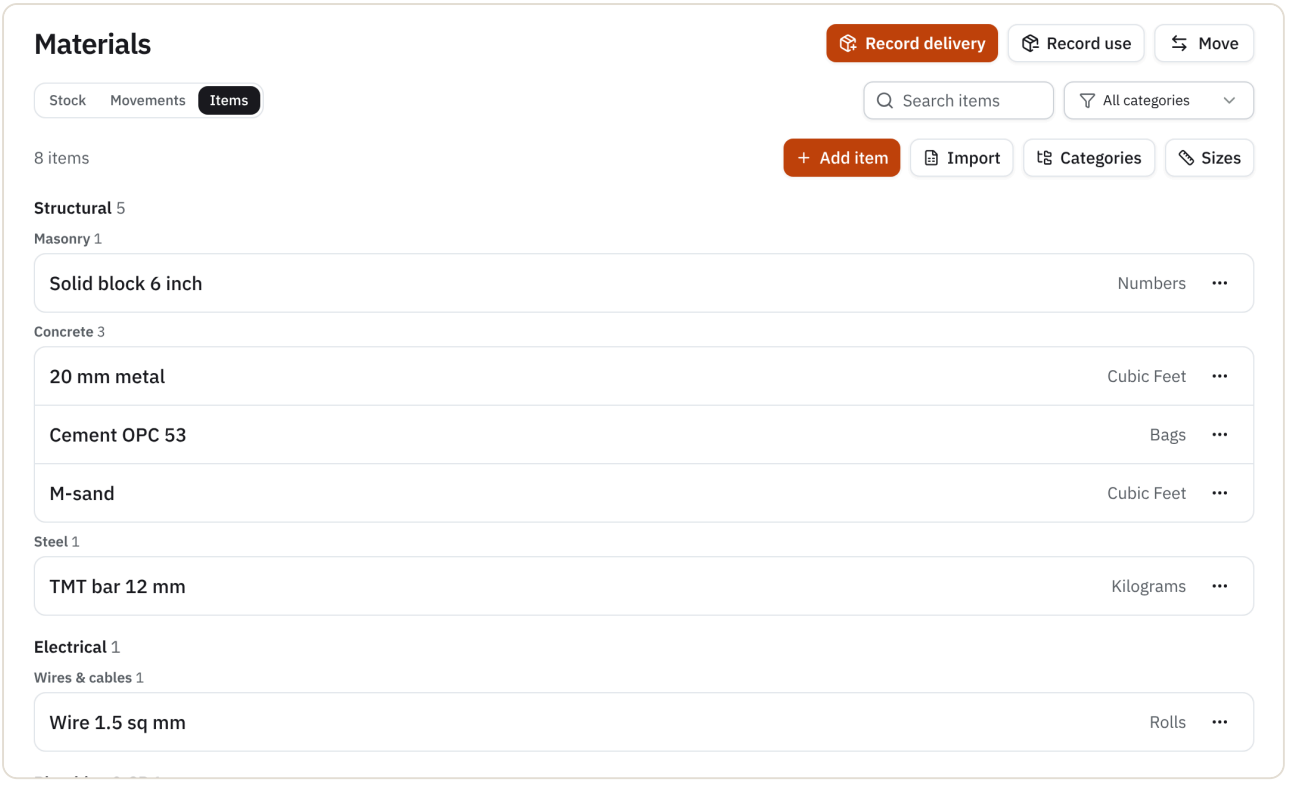
Date	Site	Material	What	Details	In	Out	Cost	By
Fri, Nov 6	Main building · Villa, Kakkanad	Wall putty	Used	Ground floor interior		−70 bags		Demo
Thu, Nov 5	Main building · Villa, Kakkanad	M-sand	Used	Ground floor walls		−180 cft		Demo
	Main building · Villa, Kakkanad	Cement OPC 53	Used	Ground floor walls		−38 bags		Demo
	Main building · Villa, Kakkanad	Solid block 6 inch	Used	Ground floor walls		−650 nos		Demo
	Main building · Villa, Kakkanad	Wall putty	Delivery	Malabar Hardware	+40 bags			Demo
	Main building · Villa, Kakkanad	Cement OPC 53	Delivery	Malabar Hardware	+100 bags		₹41,400	Demo
Wed, Nov 4	Main building · Villa, Kakkanad	Cement OPC 53	Used	Plinth beam		−60 bags		Demo
	Main building · Villa, Kakkanad	Solid block 6 inch	Delivery	Sree Blocks	+1,500 nos		₹63,000	Demo
Tue, Nov 3	Main building · Villa, Kakkanad	TMT bar 12 mm	Used	Columns, ground floor		−900 kg		Demo
	Main building · Villa, Kakkanad	TMT bar 12 mm	Delivery	Kerala Steels	+2,000 kg		₹1,36,000	Demo
Mon, Nov 2	Main building · Villa, Kakkanad	20 mm metal	Used	Column footings, grid B		−260 cft		Demo
	Main building · Villa, Kakkanad	M-sand	Used	Column footings, grid B		−210 cft		Demo
	Main building · Villa, Kakkanad	Cement OPC 53	Used	Column footings, grid B		−45 bags		Demo
	Main building · Villa, Kakkanad	20 mm metal	Delivery	Periyar Sands	+500 cft		₹23,000	Demo
Sun, Nov 1	Main building · Villa, Kakkanad	M-sand	Delivery	Periyar Sands	+600 cft		₹34,800	Demo
	Main building · Villa, Kakkanad	Cement OPC 53	Delivery	Malabar Hardware	+200 bags		₹80,000	Demo
Fri, Oct 30	Main building · Villa, Kakkanad	Wall putty	Opening stock		+25 bags			Demo

Movements tab, last 30 days. The chat icon marks entries sent on WhatsApp.

What must be on it: date, site, material, kind (delivery, use, move, opening stock, return), supplier or work, quantity in green or red, cost, who reported it and whether it came from WhatsApp; totals on top; Excel/CSV export for accounts; corrections shown as reversals, never as silent edits.

Items Working Admins and PMs

Answers: **which materials do we use, in which category, counted in which unit?**



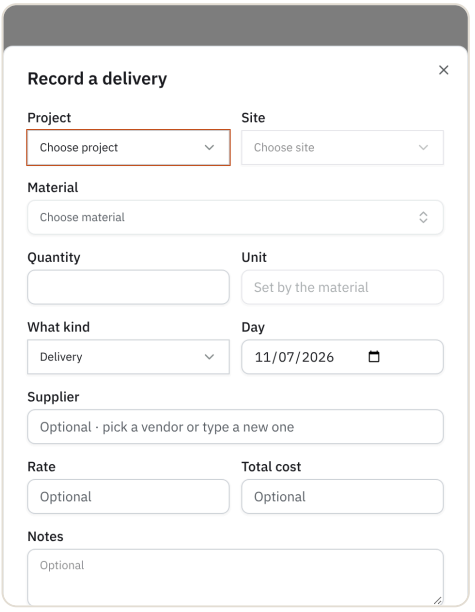
Items tab: Structural › Concrete › Cement OPC 53, counted in bags.

What must be on it: item name and its other names (aliases), category and sub-category, stock unit, sizes ("1 load = 60 cft"), import from Excel. Set up once; mostly untouched after.

Record delivery · use · move Working

For the office. Site people use WhatsApp instead.

Only **site, material and quantity** are needed. Supplier, rate and notes are optional and can come later from the bill. The unit is set by the material, so nobody types "bag" five different ways. **Record use** asks what it was used for; **Move** takes it out of one site and into another as one record.



Requests

Planned · V1 | Tracker PRC

Answers: **what does the site need, by when, and has anyone acted on it?** Comes before a purchase order (Guide 04 Procurement).

Requests — New request

NEEDED BY	SITE	MATERIAL	QTY	FOR	ASKED BY	STATUS
Fri 13 Nov	Main building	Cement OPC 53	200 bags	First-floor slab	Arun	New
Mon 16 Nov	Main building	TMT bar 12 mm	1.5 t	Slab steel	Arun	Ordered · PO-0043
Today	Chennai site	PVC pipe 1 inch	10 nos	Toilet block	Ravi	Stock at Kakkanad: move?

Must have: needed-by date, the work it is for, who asked, status through to delivery; a warning when another site already has it in stock; one tap to turn it into an order.

Reconciliation

Planned · V1 | Tracker CC-08

Answers the GM's question: **for the work planned, what should we have received and used, and what is missing?**

Reconciliation · Villa, Kakkanad — Export

MATERIAL	NEEDED (BOQ)	RECEIVED	USED	ON SITE	USED VS WORK DONE	RUNS OUT
Cement OPC 53	960 bags	700	610	90	+8% over	in 6 days
20 mm metal	3,200 cft	2,600	2,350	250	on norm	in 9 days
M-sand	4,100 cft	3,000	2,180	820	heap short 200 cft	in 14 days

Must have: what the BOQ needs (quantity × the material per unit of work), received, used, on site, whether use matches the work done, and when it runs out. This page is where Materials meets Planning and Cost control.

Central store

Planned · Later

For builders with a yard or godown: bulk stock at the store, issued to sites like a move. Same register, one more "site".

Rules we design by

Do

- Put the unit next to every number: "217 bags", never "217".
- Show days left, not just quantity. "Lasts ~10 days" is what a PM acts on.
- Show negative stock in red with a banner. A gap is information, not an error to hide.
- Let every number open the entries behind it.
- Group by category, so electrical fittings don't bury cement.
- Ask a unit size once, company-wide, from an admin.
- Mark missing prices as "Not recorded".
- Keep history: corrections are reversals with a reason.

Don't

- Make site staff open the dashboard or fill forms.
- Require supplier, rate or bill at the gate.
- Let people type units freely ("bag", "bags", "bg").
- Guess a "close" material name.
- Show stock without saying for which site and date.
- Edit old entries silently.
- Fill the screen with 20 columns; three questions, three tabs.
- Show ₹0 for "we don't know".

7 · ON WHATSAPP

Most of Materials is done in chat

Working today (each write waits for the person's Save):

Record a delivery	*"Malabar-il ninnu 200 bag cement vannu"*
Record use	*"45 bag cement footing-nu use cheythu"*
Move to another site	*"20 bag cement Chennai site-ilekku kondupoyi"*
Ask stock	*"Cement ethra und?"*
Stock for a period	*"This week cement: previous, used, balance"*
What to order	*"Enthokke theerarayi?" (what is running low)*
By category	*"Electrical stock?"*
New material	*"Wash basin 20 vannu": added to the list, with its category suggested*
Sizes	*"1 load M-sand = 60 cft" (admin or PM)*
Undo	*Take back a delivery, use or move saved by mistake*

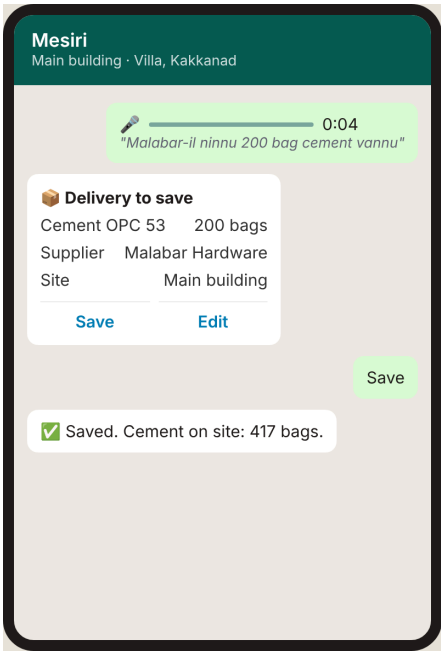


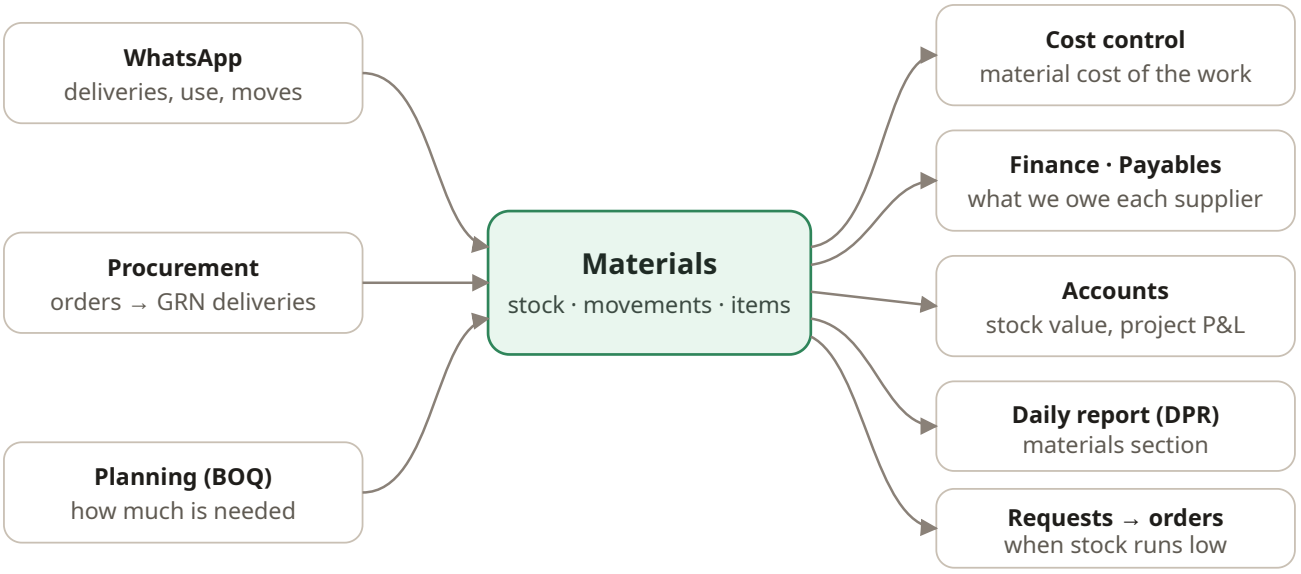
Illustration of the flow, not a screenshot. The real card's wording differs.

Planned: requests ("need 200 bags by Friday"), orders and duplicate warnings, bill photo matched to the delivery, a daily nudge when a site reports deliveries but no use.

How it decides: the code, not the AI, checks units and material names. The AI never invents a size ("one truck" is not a size) and never saves a material that is only "near" what was said.

8 · HOW IT CONNECTS TO EVERYTHING ELSE

Materials sits in the middle of the money



- **In:** site messages; deliveries against purchase orders (Procurement); how much the BOQ needs (Planning).
- **Out:** the material cost of each day's work (Cost control); what is owed to each supplier (Payables); stock value and project profit (Accounts); the materials part of the daily report; new requests when stock runs low.

9 · MESIRI VS OTHER SOFTWARE

TYPICAL CONSTRUCTION APP OR ERP	MESIRI
The storekeeper installs an app, logs in, fills a GRN form.	One WhatsApp line or voice note, in Malayalam, Manglish or English.
A register of entries; someone has to read it.	Answers on top: running low, days left, spend by category; planned: used vs work done.
Weeks of setup with item codes.	Standard builder categories built in; the customer's item list imported from Excel.

10 · QUESTIONS CUSTOMERS WILL ASK

"Can it handle electrical and plumbing detail, fittings, bends, wire lengths?"

Yes. Each size or type is its own item under a category (Electrical › Conduits, Plumbing › Pipe fittings). Mesiri suggests the category when a new item is added.

"What if the engineer forgets to report use?"

Then stock stays too high, and we must be honest about it. Today the daily report shows what each site sent; planned reconciliation against the BOQ will flag a site whose use doesn't match its work.

"We measure sand in loads, they bill in cft."

Each item has one stock unit. Mesiri converts known units itself and asks once for a load's size.

"Can we see all sites together?"

Yes: All projects in the top bar, or one project or site. Moves between sites are one record.

"Who can change numbers?"

Nobody edits old entries. A correction reverses the old one, with who and why, so history is never lost.

"We already have our item list in Excel."

Items › Import loads it, with categories and opening stock.

11 · KEY TERMS

Delivery / GRN	Material that arrived. A GRN (goods receipt note) is the formal record of it, ideally against an order.
Challan	The delivery note that comes with the lorry.
Opening stock	What was already on site when counting started.
Use / consumption	Material that went into the work.
Stock unit	The one unit an item is counted in (bags, cft, kg, nos).
Conversion / size	How many stock units are in another unit ("1 load = 60 cft").
Move / transfer	Material sent from one site to another.
Wastage	Material lost or spoiled (hardened cement, broken blocks).
Reconciliation	Comparing what should have been used (by the BOQ) with what was.
CFT, cum, brass	Volumes: 1 cum ≈ 35.3 cft; 1 brass = 100 cft.

CHECK YOURSELF

1. Before 60, received 300, used 143. What is in stock, and how do you check it on the Stock tab?
2. Why is wall putty showing −5, and what are the two possible reasons?
3. Which tab answers "what happened on Tuesday at the villa?"
4. A lorry brings "one load" of M-sand. What does Mesiri need before it can add it to stock in cft?
5. Which planned page answers the GM's "4 loads missing at closeout" problem, and what does it need from Planning?

Answers: 1. 217; the row shows Before, Received, Used, In stock side by side. 2. More was reported used (70) than ever arrived (65): a delivery wasn't recorded, or use was over-reported. 3. Movements, filtered to that site and date. 4. The size of a load in cft, answered once by an admin or PM. 5. Reconciliation; it needs the BOQ quantities and the material per unit of work.